



Natural Language Processing

Lecture

**Evaluating Hypothesis
(Models)**

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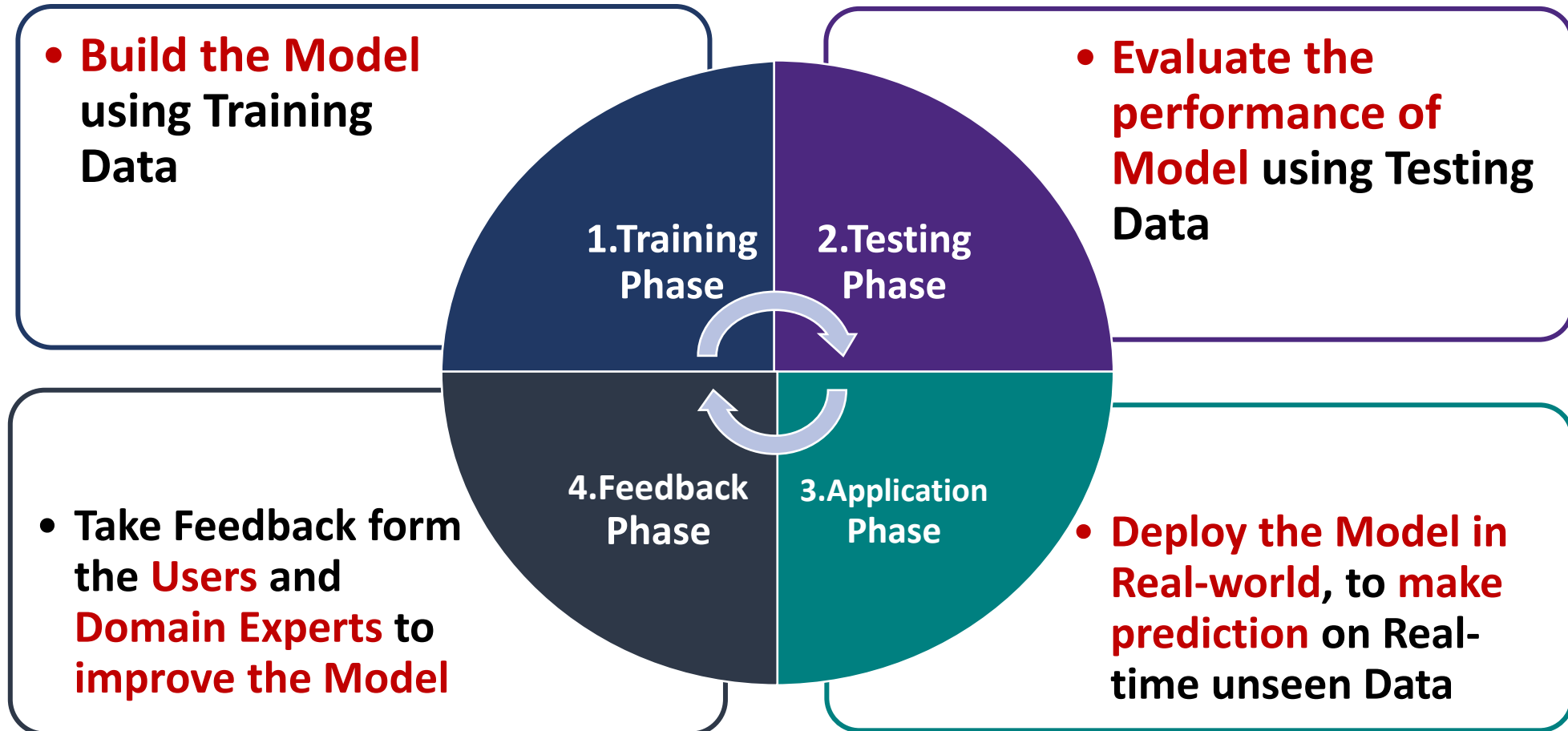
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Recall The Machine Learning Cycle

Four phases of a Machine Learning Cycle are



Lecture Focus

A **Model is evaluated** in **Testing Phase** using Testing Data

Evaluation is **quantified** using
☐ Evaluation Measures(s)

Therefore, this Lecture will **only focus** on
☐ Testing Phase of Machine Learning Cycle

Why Evaluate Hypotheses (Model)?



Why Evaluate Hypothesis (Model)? (Cont.)

Question

- ❑ Why should we **evaluate h (Model)**?

Answer

- The **main goal** of **building a Model** (Training / Learning) is to **use** it in doing **Real-world Tasks** with **good Accuracy**
- No one can **perfectly predict**, that if a Model performs well in **Training Phase**, then it will also perform well in **Real-world**
- However, **before deploying a Model** in the Real-world, it is **important** to know
 - How **well** will it **perform** on **unseen** Real-time Data?

Evaluation Measures for Classification Problems



Evaluation Measures for Classification Problem

- Some of the **most popular** and **widely used** Evaluation Measures for Classification Problems are
 - Accuracy
 - Precision
 - Recall
 - F_1
 - Area Under the Curve (AUC)

Accuracy

Definition

- Accuracy is defined as the proportion of correctly classified Test instances

Formula

$$\text{Accuracy} = \frac{\text{Number of Correctly Classified Test Instances}}{\text{Total Number of Test Instances}}$$

Note

$$\text{Accuracy} = 1 - \text{Error}$$

Accuracy (Cont.)

ID	Text	Actual Class	Predicted Class
1	Community sports event organized in the city park	Non-Terrorism	Non-Terrorism
2	Students attended a technology workshop at university	Non-Terrorism	Non-Terrorism
3	Local market announced seasonal discount offers	Non-Terrorism	Non-Terrorism
4	Suspicious online post discussing violent attack plans	Non-Terrorism	Terrorism
5	Extremist group released threatening statements online	Terrorism	Non-Terrorism
6	Illegal weapons training camp discovered in remote area	Terrorism	Non-Terrorism
7	Terror organization claimed responsibility for explosion	Terrorism	Non-Terrorism
8	Security forces arrested members planning a bombing	Terrorism	Terrorism
9	Intelligence agencies detected terrorist communication network	Terrorism	Terrorism
10	Authorities prevented a planned terror attack in the capital	Terrorism	Terrorism

$$\text{Accuracy} = \frac{\text{Number of Correctly Classified Test Instances}}{\text{Total Number of Test Instances}}$$

$$\text{Accuracy} = \frac{6}{10}$$

$$\text{Accuracy} = 0.6$$

Accuracy (Cont.)

Question

- When it will **more suitable** to use Accuracy measure?

Answer

- Accuracy evaluation measure is **more suitable** to use for **evaluation** of Machine Learning Algorithms when we have
 - ❑ Balanced Data

Strengths and Weaknesses

- We can **evaluate** and **compare** various Machine Learning Algorithms using Accuracy evaluation measure
- Accuracy fails to **accurately** evaluate a Machine Learning Algorithm when Test Data is **highly unbalanced**

Example – Calculating Accuracy



Problem Description

Rasheed wants to evaluate his proposed Machine Learning Algorithms (called Excellent Learner) for the task of counter terrorism Identification. His Test Data comprises of 600 instances (300 Terrorism and 300 Non-Terrorism). Rasheed applied Excellent Learner on Test Data and 500 instances were correctly classified. What is the Accuracy?

Example – Calculating Accuracy (Cont.)

Calculating Accuracy for Efficient Learner (Machine Learning Algorithm)

$$\text{Accuracy} = \frac{\text{Number of Correctly Classified Test Instances}}{\text{Total Number of Test Instances}}$$

$$\text{Accuracy} = \frac{500}{600}$$

$$\text{Accuracy} = 0.83$$

Example – Accuracy is a Poor Measure for Highly Unbalanced Data

- ❑ Consider a Binary Classification Problem with two classes: Positive and Negative. Test Data comprises of 1000 instances, out of which 995 instances are Negative and 5 are Positive.
 - ❑ A Machine Learning Algorithm which **always predicts** Negative, will have an Accuracy of 0.995
-
- ❑ Problem
 - Machine Learning Algorithm has **very high** Accuracy (99.5%) on Test Data, even though it **never correctly predicts Positive Test Examples**
-
- ❑ A Possible Solution
 - Confusion Matrix

Confusion Matrix

Definition

A Confusion Matrix is a table used to describe the **performance of a Classification Model** (or Classifier) on a Set of Test Examples (Test Data), whose **Actual Values** (or True Values) are **known**

Purpose

- ❑ To get **deeper insights** into Model / Classifier **behavior**

Advantages

- ❑ Confusion Matrix **allows** us to **visualize** the **performance** of a Model / Classifier
- ❑ Confusion Matrix **allows** to **separately** get **insights** into the **Errors** made by **each Class**
- ❑ Confusion Matrix gives insights to **both**

Confusion Matrix (Cont.)

Evaluation Measures

- ❑ Confusion Matrix **allows** us to **compute** many **different Evaluation Measures** including:

- Accuracy
- Precision
- True Positive Rate (or Recall)
- True Negative Rate
- False Negative Rate
- False Positive Rate
- F_1

Confusion Matrix

Confusion Matrix for a Machine Learning Problem with **n** Number of Classes is given below,

		Predicted				
		Class 01	Class 02		Class n-1	Class n
Actual	Class 01					
	Class 02					
						
	Class n-1					
	Class n					

Confusion Matrix for Concept Learning

Confusion Matrix for Concept Learning (a.k.a. Binary Classification Problem) is given below,

		Predicted	
		Class 01	Class 02
Actual	Class 01	a	b
	Class 02	c	d

Confusion Matrix for Concept Learning (Cont.)

❑ Considering that

- Class 01 = Non-Terrorism
- Class 02 = Terrorism

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Extracting Various Evaluation Measures from Confusion Matrix

□ Next Slides I will try to explain how to extract the following Evaluation Measures from Confusion Matrix

- Accuracy
- Precision
- True Positive Rate (or Recall)
- True Negative Rate
- False Negative Rate
- False Positive Rate
- F_1

Extracting Various Evaluation Measures from Confusion Matrix (Cont.)

Accuracy

Accuracy (AC) is the proportion of the total number of predictions that were correct

$$AC = \frac{(TP) + (TN)}{(TP) + (TN) + (FP) + (FN)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

❑ This is the same as Accuracy evaluation measure defined earlier

Evaluation Measures from Confusion Matrix (Cont.)

Recall or True Positive Rate (TPR) or Sensitivity

Recall or True Positive Rate (TPR) or Sensitivity is the proportion of Positive cases that were correctly classified

$$TPR = \frac{(TP)}{(FN) + (TP)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Evaluation Measures from Confusion Matrix (Cont.)

True Negative Rate or Specificity

True Negative Rate (TNR) or Specificity is defined as the proportion of Negatives cases that were classified correctly

$$TNR = \frac{a(TN)}{a(TN) + b(FP)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Evaluation Measures from Confusion Matrix (Cont.)

False Positive Rate

False Positive Rate (FPR) is the proportion of Negative cases that were incorrectly classified as Positive

$$FPR = \frac{(FP)}{(FP) + (TN)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Extracting Various Evaluation Measures from Confusion Matrix (Cont.)

False Negative Rate

False Negative Rate (FNR) is the proportion of Positive cases that were incorrectly classified as Negative

$$FNR = \frac{(FN)}{(FN) + (TP)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Extracting Various Evaluation Measures from Confusion Matrix (Cont.)

Precision

Precision (P) is the proportion of the predicted Positive cases that were correct

$$P = \frac{d(TP)}{b(FP) + d(TP)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Trade Off Between Precision and Recall

Problem

- There is a **trade-off** between Precision and Recall

A Possible Solution

- Find out Evaluation Measures which **efficiently combine** Precision and Recall
- Example
 - F-measure (combines Precision and Recall)

F₁-measure

F₁-measure

When we assign **same weights** to Precision and Recall i.e. $\beta = 1$, the F-measure becomes F₁-measure

$$F_1 = \frac{2 * Precision * Recall}{Precision + Recall}$$

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



Dataset

ID	Text	Actual Class	Predicted Class
1	Community sports event organized in the city park	Non-Terrorism	Non-Terrorism
2	Students attended a technology workshop at university	Non-Terrorism	Non-Terrorism
3	Local market announced seasonal discount offers	Non-Terrorism	Non-Terrorism
4	Suspicious online post discussing violent attack plans	Non-Terrorism	Terrorism
5	Extremist group released threatening statements online	Terrorism	Non-Terrorism
6	Illegal weapons training camp discovered in remote area	Terrorism	Non-Terrorism
7	Terror organization claimed responsibility for explosion	Terrorism	Non-Terrorism
8	Security forces arrested members planning a bombing	Terrorism	Terrorism
9	Intelligence agencies detected terrorist communication network	Terrorism	Terrorism
10	Authorities prevented a planned terror attack in the capital	Terrorism	Terrorism

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



Confusion Matrix

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	3	1
	Terrorism	3	3

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



Accuracy

$$AC = \frac{TP + TN}{TP + FN + FP + TN}$$

$$AC = \frac{3 + 3}{3 + 1 + 3 + 3}$$

$$AC = \frac{6}{10}$$

$$AC = 0.6$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 3	FN 1
	Terrorism	FP 3	TN 3

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



True Positive Rate (TPR) or Recall or Sensitivity

$$TPR = \frac{TP}{TP + FN}$$

$$TPR = \frac{3}{3 + 1}$$

$$TPR = \frac{3}{4}$$

$$TPR = 0.75$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 3	FN 1
	Terrorism	FP 3	TN 3

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



True Negative Rate (TNR) or Specificity

$$TNR = \frac{TN}{TN + FP}$$

$$TNR = \frac{3}{3 + 3}$$

$$TNR = \frac{3}{6}$$

$$TNR = 0.50$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 3	FN 1
	Terrorism	FP 3	TN 3

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



False Positive Rate (FPR)

$$FPR = \frac{FP}{FP + TN}$$

$$FPR = \frac{3}{3 + 3}$$

$$FPR = \frac{3}{6}$$

$$FPR = 0.50$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 3	FN 1
	Terrorism	FP 3	TN 3

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



False Negative Rate (FNR)

$$FNR = \frac{FN}{FN + TP}$$

$$FNR = \frac{1}{1 + 3}$$

$$FNR = \frac{1}{4}$$

$$FNR = 0.25$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 3	FN 1
	Terrorism	FP 3	TN 3

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



Precision

$$P = \frac{TP}{TP + FP}$$

$$P = \frac{3}{3 + 3}$$

$$P = \frac{3}{6}$$

$$P = 0.50$$



F₁ Measure

$$F_1 = \frac{2 * Precision * Recall}{Precision + Recall}$$

$$F_1 = \frac{2 * 0.5 * 0.75}{0.5 + 0.75}$$

$$F_1 = \frac{0.75}{1.25} = 0.6$$

Example 02 – Confusion Matrix for Ternary Classification Problem



□ Consider the following Machine Learning Problem

Irfan had a collection of 2000 documents, which can be categorized as terrorism (1000 documents), Non-Terrorism (600 documents) and Neutral (400 documents). He applied a Naïve Bayes classifier on it. Out of 1000 terrorism documents, half were correctly classified and 300 were classified as Neutral. For Non-Terrorism class, 150 were classified as terrorism and 250 as Non-Terrorism. In case of Neutral class, 300 were correctly classified and 50 were classified as Non-Terrorism.

Example 02 – Confusion Matrix for Ternary Classification Problem (Cont.)



Confusion Matrix

❑ Total Documents = 2000

❑ Terrorism Documents = 1000

❑ Non-terrorism Documents = 600

❑ Neutral Documents = 400

		Predicted		
		Terrorism	Non-terrorism	Neutral
Actual	Terrorism	500	200	300
	Non-terrorism	150	250	200
	Neutral	50	50	300

Example 02 – Confusion Matrix for Ternary Classification Problem (Cont.)



Confusion Matrix

		Predicted		
		Terrorism	Non- terrorism	Neutral
Actual	Terrorism	a	b	c
	Non- terrorism	d	e	f
	Neutral	g	h	i

Example 02 – Confusion Matrix for Ternary Classification Problem (Cont.)



Accuracy

$$AC = \frac{a + e + i}{a + b + c + d + e + f + g + h + i}$$

$$AC = \frac{500 + 250 + 300}{500 + 200 + 300 + 150 + 250 + 200 + 50 + 50 + 300}$$

$$AC = \frac{1050}{2000}$$

$$AC = 0.525$$

Example 02 – Confusion Matrix for Ternary Classification Problem (Cont.)



Confusion Matrix

		Predicted		
		Terrorism	Non-terrorism	Neutral
Actual	Terrorism	500	200	300
	Non-terrorism	150	250	200
	Neutral	50	50	300

Terrorism Class (Support = $500 + 200 + 300 = 1000$)

Metric	Formula	Calculation	Result
True Positive Rate (Recall)	$TP / (TP + FN)$	$500 / (500 + 500)$	0.5000
True Negative Rate	$TN / (TN + FP)$	$800 / (800 + 200)$	0.8000
False Positive Rate	$FP / (FP + TN)$	$200 / (200 + 800)$	0.2000
False Negative Rate	$FN / (FN + TP)$	$500 / (500 + 500)$	0.5000
Precision	$TP / (TP + FP)$	$500 / (500 + 200)$	0.7143

Example 02 – Confusion Matrix for Ternary Classification Problem (Cont.)



Confusion Matrix

		Predicted		
		Terrorism	Non-terrorism	Neutral
Actual	Terrorism	500	200	300
	Non-terrorism	150	250	200
	Neutral	50	50	300

Non-Terrorism Class (Support = 150 + 250 + 200 = 600)

Metric	Formula	Calculation	Result
True Positive Rate (Recall)	$TP / (TP + FN)$	$250 / (250 + 350)$	0.4167
True Negative Rate	$TN / (TN + FP)$	$1150 / (1150 + 250)$	0.8214
False Positive Rate	$FP / (FP + TN)$	$250 / (250 + 1150)$	0.1786
False Negative Rate	$FN / (FN + TP)$	$350 / (350 + 250)$	0.5833
Precision	$TP / (TP + FP)$	$250 / (250 + 250)$	0.5000

Example 02 – Confusion Matrix for Ternary Classification Problem (Cont.)



Confusion Matrix

		Predicted		
		Terrorism	Non-terrorism	Neutral
Actual	Terrorism	500	200	300
	Non-terrorism	150	250	200
	Neutral	50	50	300

Neutral Class (Support = 50 + 50 + 300 = 400)

Metric	Formula	Calculation	Result
True Positive Rate (Recall)	$TP / (TP + FN)$	$300 / (300 + 100)$	0.7500
True Negative Rate	$TN / (TN + FP)$	$1100 / (1100 + 500)$	0.6875
False Positive Rate	$FP / (FP + TN)$	$500 / (500 + 1100)$	0.3125
False Negative Rate	$FN / (FN + TP)$	$100 / (100 + 300)$	0.2500
Precision	$TP / (TP + FP)$	$300 / (300 + 500)$	0.3750

Example 02 – Confusion Matrix for Ternary Classification Problem (Cont.)



Classification Report

Formula: $Macro\ Average = \frac{x_1 + x_2 + x_3}{3}$

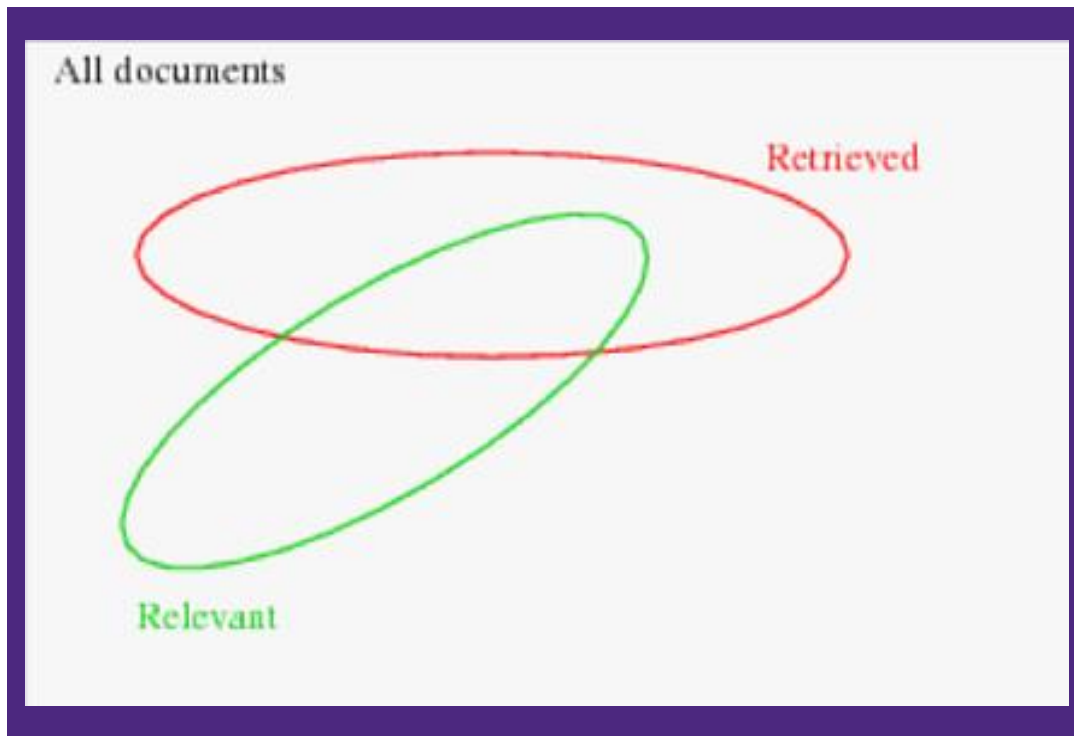
Formula: $Weighted\ Average = \frac{\sum (Metric_i \times Support_i)}{Total\ Support}$

Metric	Terror	Non-Terror	Neutral	Macro Avg Formula	Macro Avg Result	Weighted Avg Formula	Weighted Avg Result
True Positive Rate (Recall)	0.5000	0.4167	0.7500	$((0.5000 + 0.4167 + 0.7500)/3)$	0.5556	$((0.5000 \times 1000) + (0.4167 \times 600) + (0.7500 \times 400))/2000$	0.5250
True Negative Rate	0.8000	0.8214	0.6875	$((0.8000 + 0.8214 + 0.6875)/3)$	0.7696	$((0.8000 \times 1000) + (0.8214 \times 600) + (0.6875 \times 400))/2000$	0.7839
False Positive Rate	0.2000	0.1786	0.3125	$((0.2000 + 0.1786 + 0.3125)/3)$	0.2304	$((0.2000 \times 1000) + (0.1786 \times 600) + (0.3125 \times 400))/2000$	0.2161
False Negative Rate	0.5000	0.5833	0.2500	$((0.5000 + 0.5833 + 0.2500)/3)$	0.4444	$((0.5000 \times 1000) + (0.5833 \times 600) + (0.2500 \times 400))/2000$	0.4750
Precision	0.7143	0.5000	0.3750	$((0.7143 + 0.5000 + 0.3750)/3)$	0.5298	$((0.7143 \times 1000) + (0.5000 \times 600) + (0.3750 \times 400))/2000$	0.5822
Support	1000	600	400	—	—	$(1000 + 600 + 400)$	2000

Retrieved vs. Relevant Documents



Precision and Recall address the relation between the retrieved and relevant sets of documents

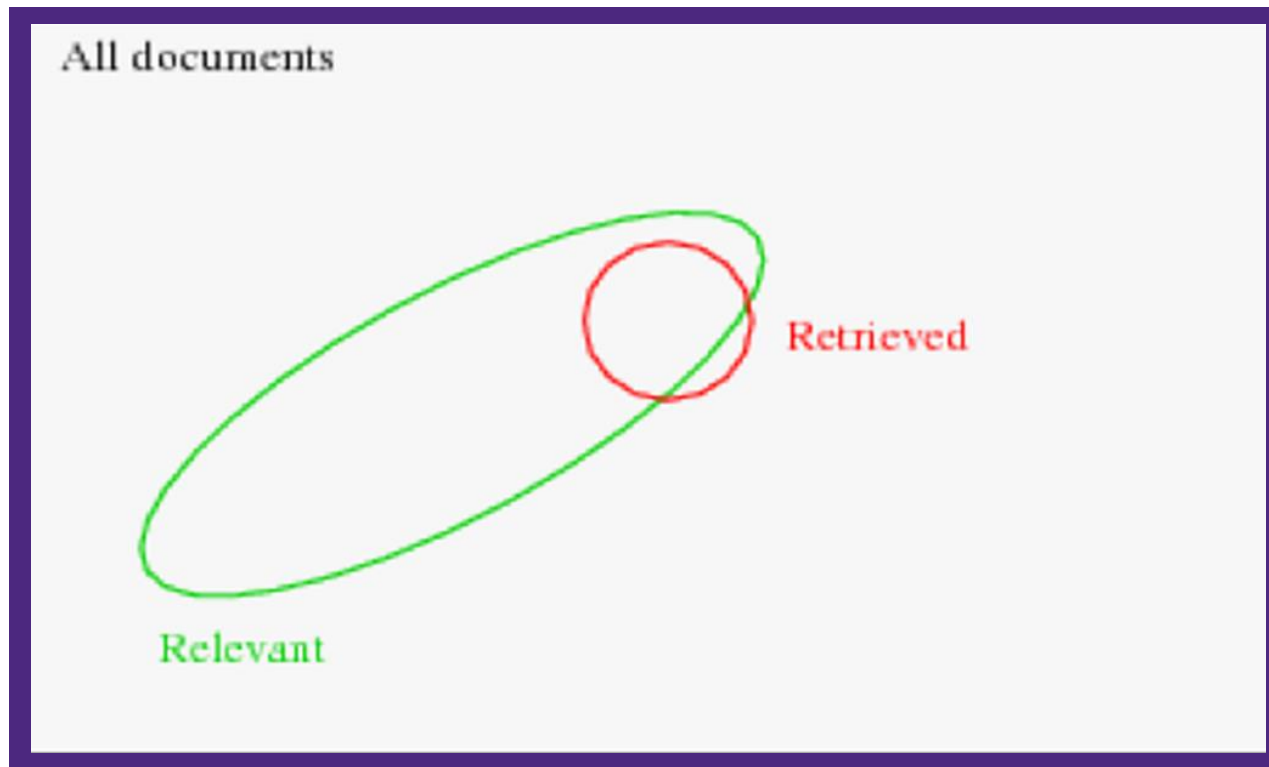


Various situations that arise can be pictorially represented in these terms

Retrieved vs. Relevant Documents



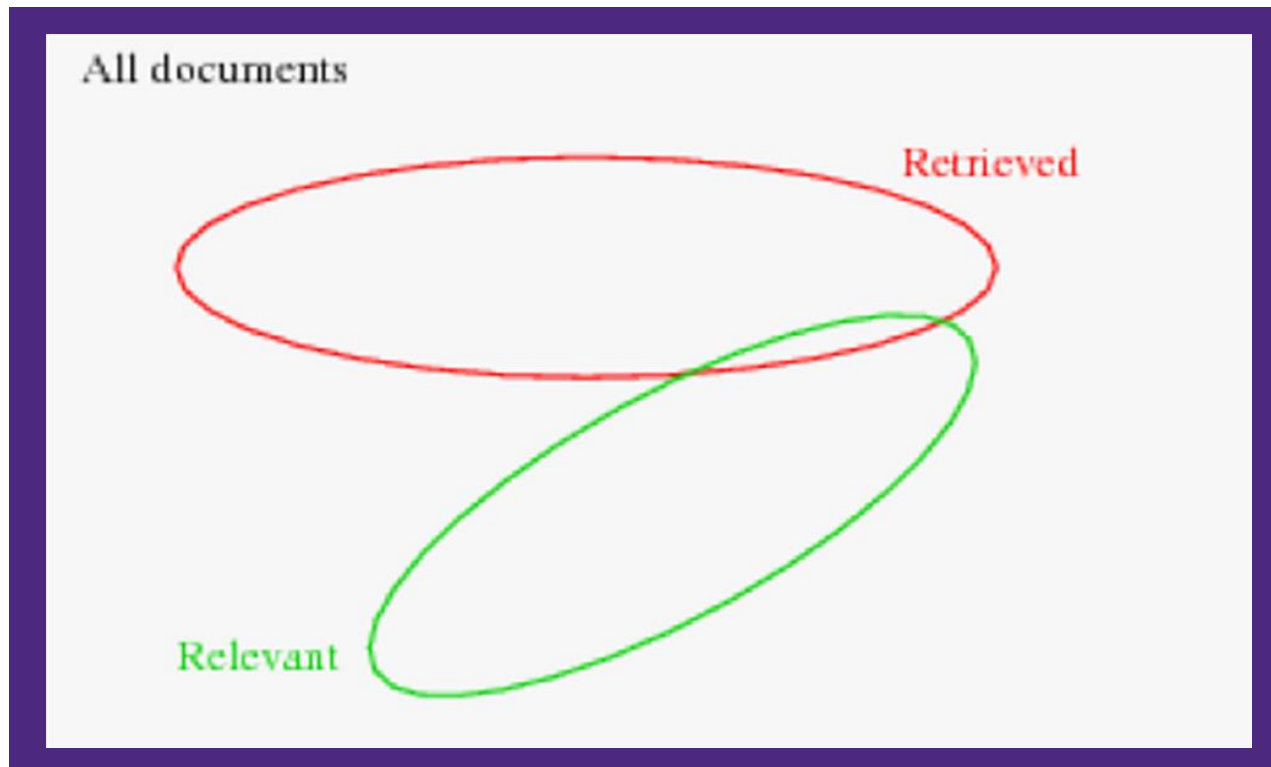
High precision, low recall



Retrieved vs. Relevant Documents



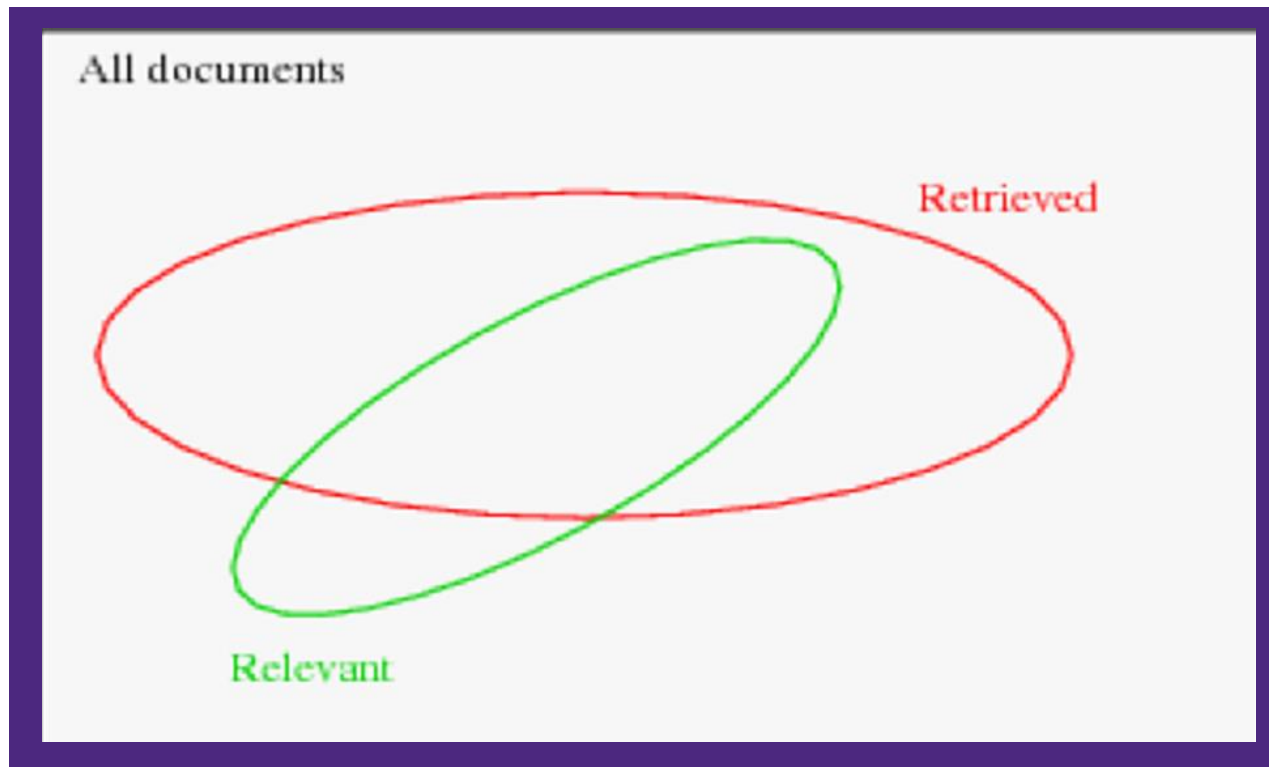
Low precision, low recall



Retrieved vs. Relevant Documents



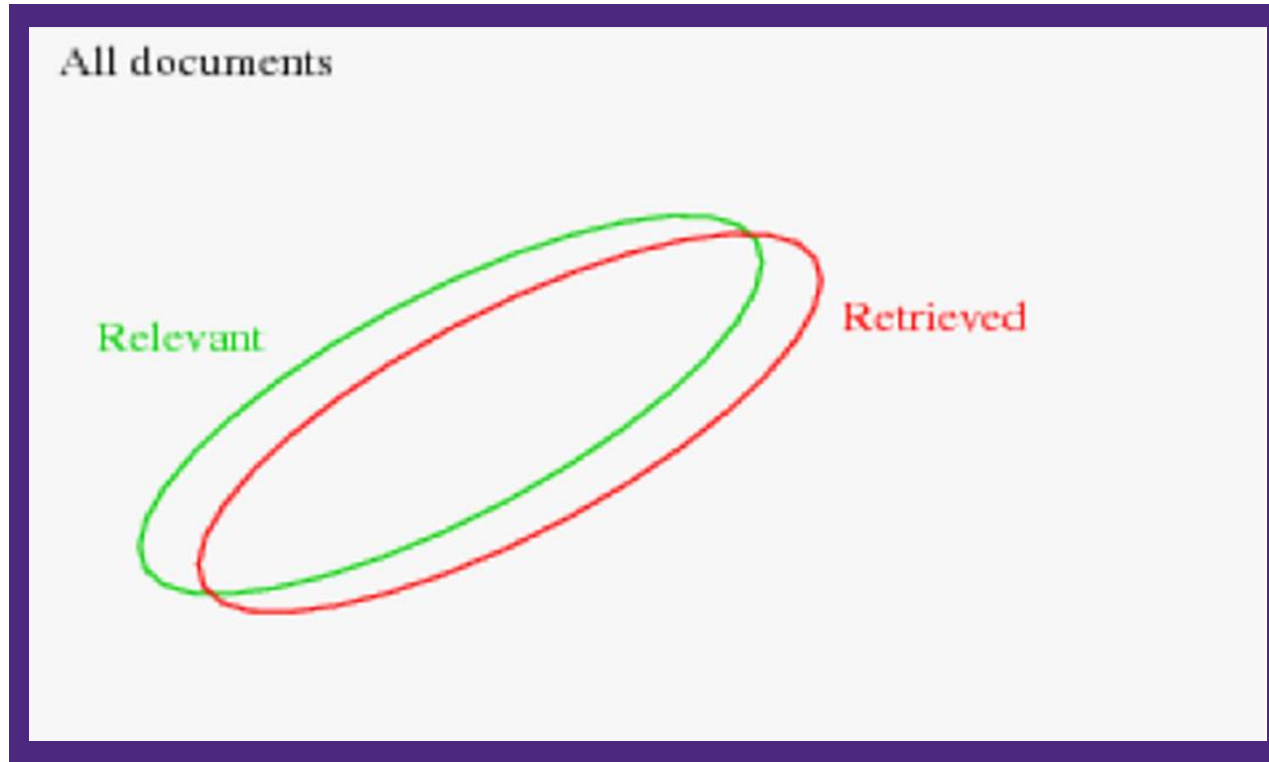
Low precision, high recall



Retrieved vs. Relevant Documents



High precision, high recall



ROC Curve



Definition

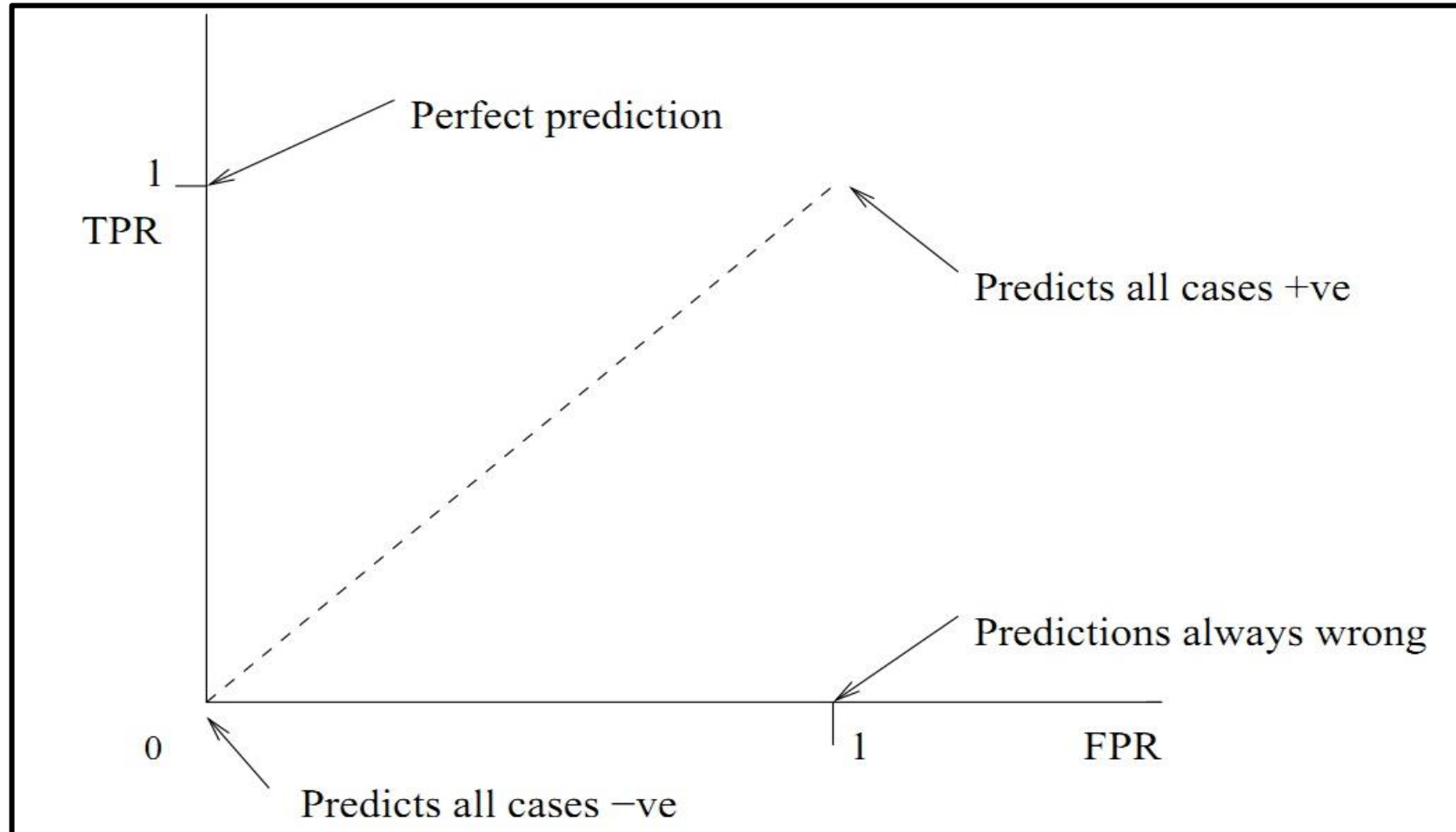
ROC Curve **summarizes** the **trade-off** between the True Positive Rate (TPR) and False Positive Rate (FPR) for a Classifier / Model using **different** Probability Thresholds



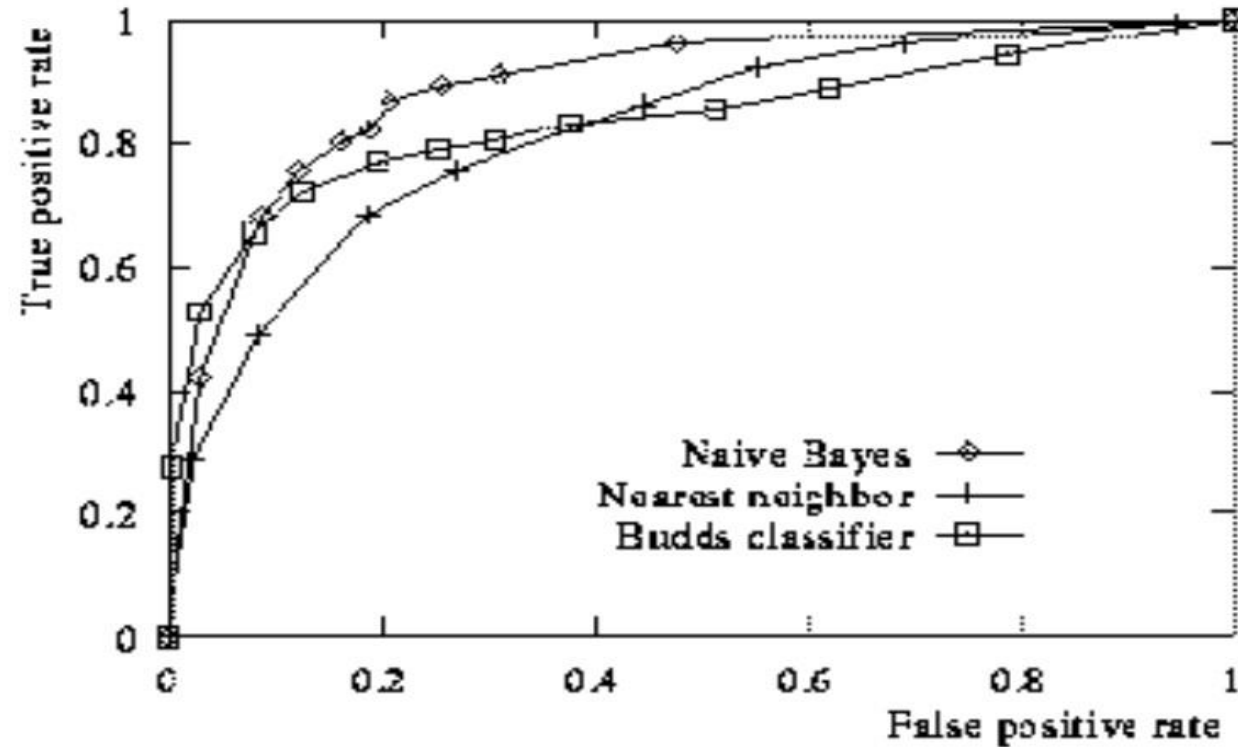
Suitable to Use

- ☐ ROC Curves are **more suitable** to use when we have
 - **Balanced Data**

ROC Curve (Cont.)



ROC Curve – Example



- ROC curves for several learning algorithms at different cost trade-offs for the task of rooftop detection in aerial images (from <http://www-csli.stanford.edu/icml2k/craft.html>)

Precision-Recall Curve

Definition

Precision-Recall Curve **summarizes** the **trade-off** between Precision and Recall (or True Positive Rate) for a Classifier / Model using **different** Probability Thresholds

Suitable to Use

- ☐ Precision-Recall Curves are **more suitable** to use when we have
 - **Highly** Unbalanced Data

Area Under the Curve (AUC)



Definition

- Area Under the ROC Curve (AUC) is defined as the **probability** that randomly chosen Positive instance is ranked above than the randomly selected Negative one



Purpose

- AUC measures the **degree of separability** between Classes
 - i.e. AUC tells how much Model is capable of **distinguishing between Classes** ?

Are Under the Curve (AUC) (Cont.)



How AUC Score is calculated?

- AUC score is computed using True Positive Rate (TPR) and False Positive Rate (FPR)



Range of AUC Score

- Range of AUC Score is [0 - 1]
 - 0 means All Predictions of Model are Wrong
 - 1 means All Predictions of Model are Correct

Are Under the Curve (AUC) (Cont.)



What is a good AUC Score?

- **AUC Score = 0.5**
 - suggests **No Discrimination** (i.e., Model **do not** have the ability to **differentiate** Positive instances from Negative ones)
- **AUC Score = 0.7 to 0.8**
 - considered Acceptable
- **AUC Score = 0.8 to 0.9**
 - considered Excellent
- **AUC Score = Greater Than 0.9**
 - considered Outstanding

ROC Graph vs AUC

ROC Graph

- ❑ ROC Graph is a **probability curve**

AUC

- ❑ AUC represents **degree of separability** between Classes